

: Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment

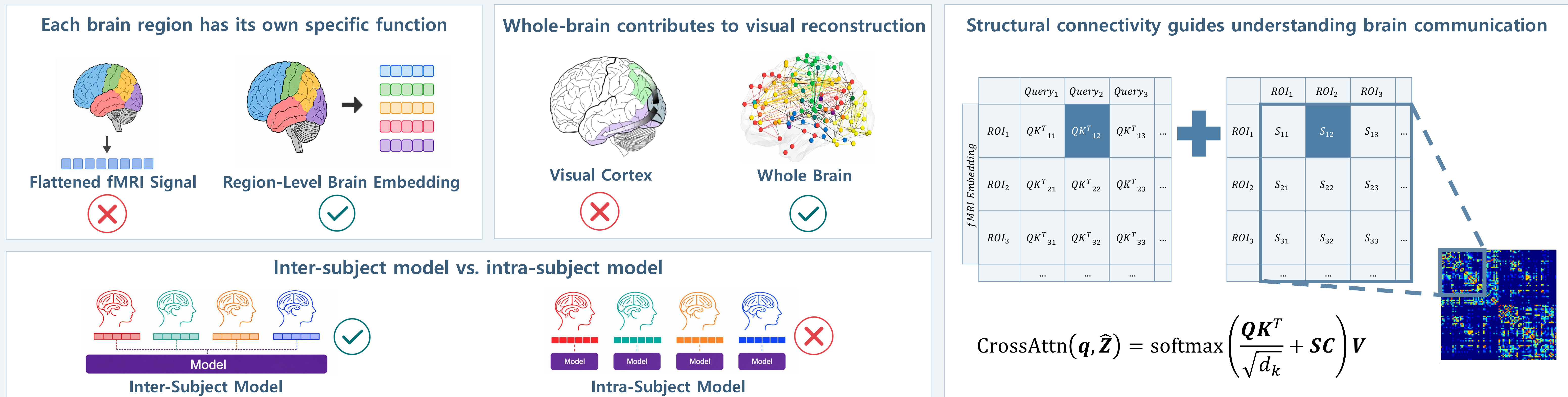
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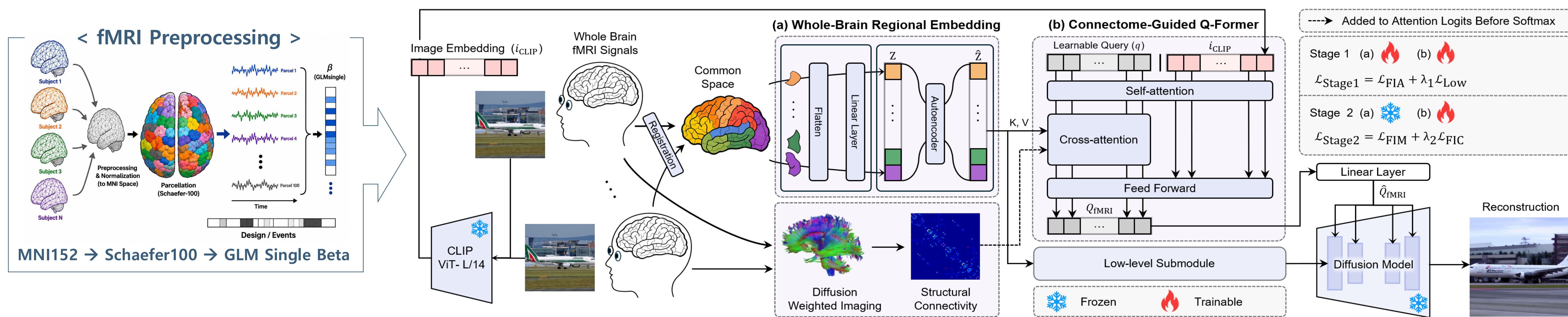
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Motivation: Can subject-invariant fMRI representations enable visual reconstruction across individuals?



Solution: ConnecToMind2

: Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment



Training Strategy

Stage	Epoch	Loss	Trained Modules
Stage 1	Epochs 0-74	FIR	Whole Brain Regional Embedding Module + Connectome-Guided Q-Former Module
		Low	Whole Brain Regional Embedding Module + Low-Level Submodule
Stage 2	Epochs 75-149	FIC FIM	Connectome-Guided Q-Former Module

FIR & Low Loss Functions

$$\text{FIR Loss MSE}(\text{CLIP-Aligned fMRI Embeddings} - \text{CLIP Image Embeddings})^2$$

$fMRI_1$	B	257	768
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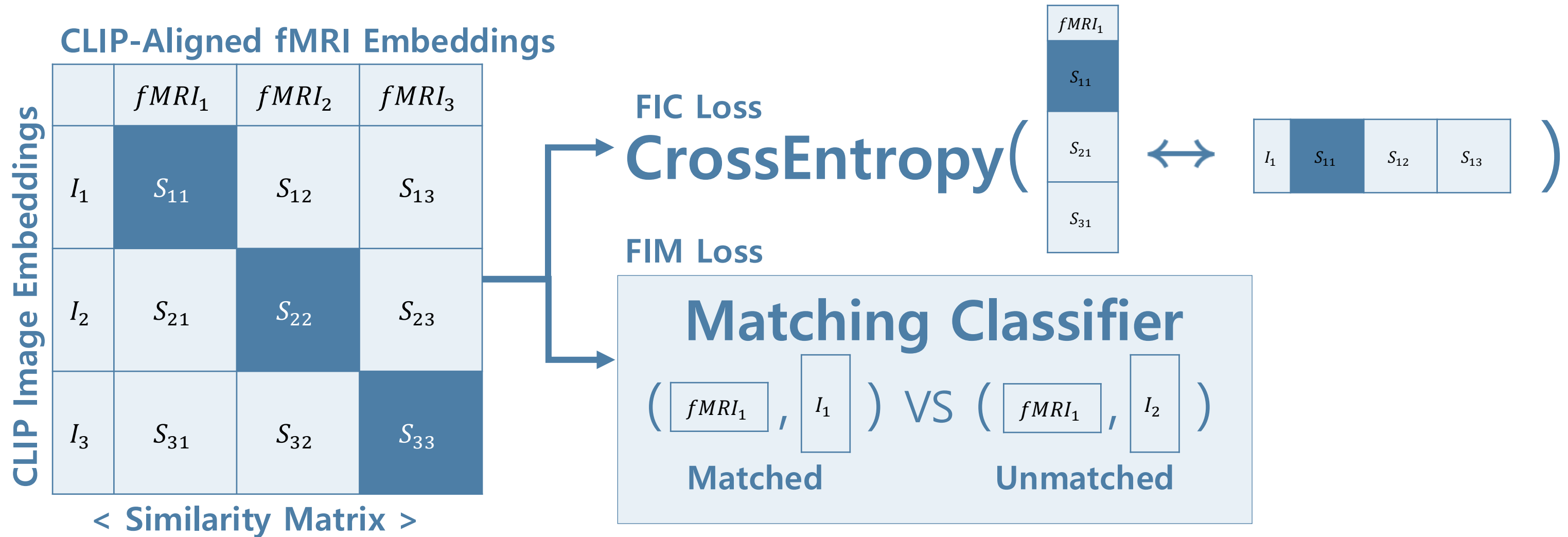
$Image_1$	B	257	768
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$$\text{Low Loss L1}(\text{SD Latent fMRI Embeddings} - \text{SD Latent Image Embeddings})$$

$fMRI_1$	B	4	64	64
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$Image_1$	B	4	64	64
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FIC & FIM Loss Functions



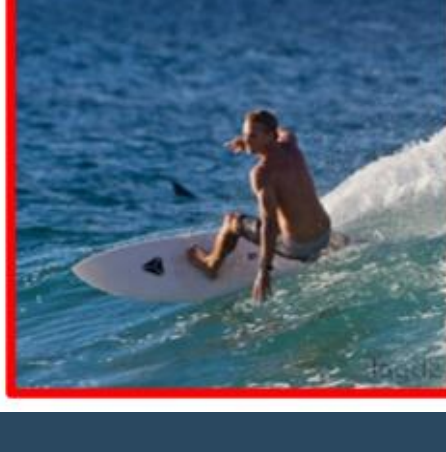
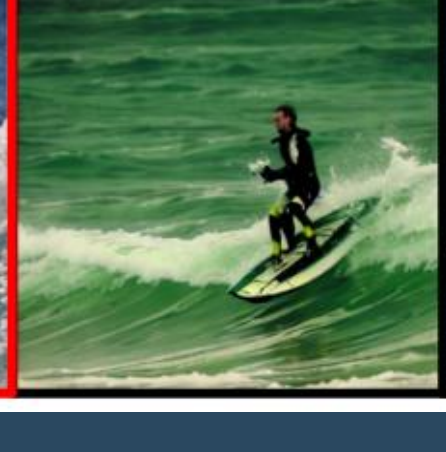
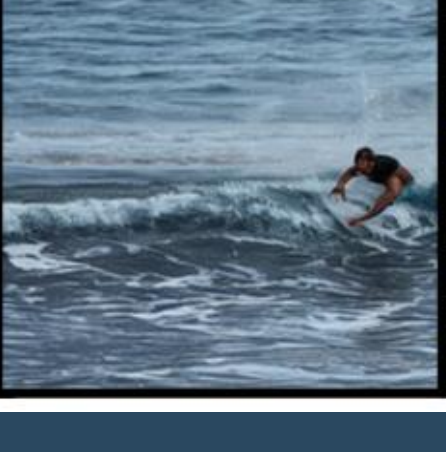
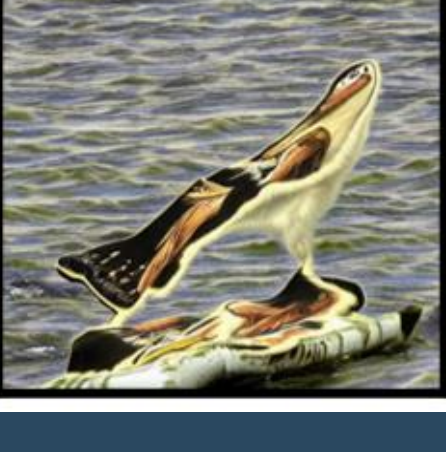
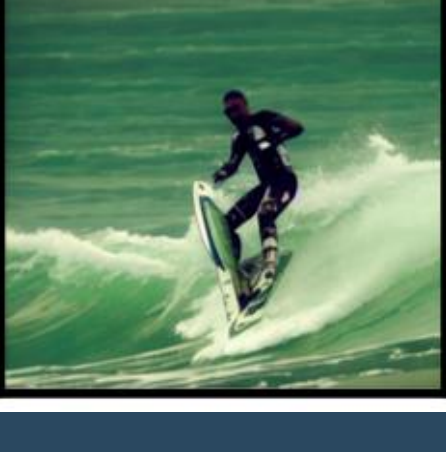
Dataset

Natural Scenes Dataset(NSD)

- Subjects : sub-01 / sub-02 / sub-05 / sub-07
- Subjects who completed all 40 sessions were chosen.
- 10,000 MS COCO Images
- Train(unshared) 9,000 / Test(shared) 1,000

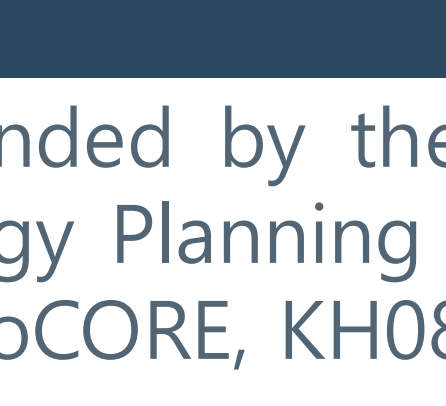
Reconstructed Image Evaluation

Regions	Model	Low-Level				High-Level			
		PCorr ↑	SSIM ↑	Alex(2) ↑	Alex(5) ↑	Incep ↑	CLIP ↑	Eff ↓	SwAV ↓
Intra	Visual MindEye1	0.167	0.290	88.5%	95.4%	92.0%	91.0%	0.686	0.402
	Visual ConnecToMind	0.170	0.297	88.5%	95.9%	93.5%	92.3%	0.666	0.388
	Whole ConnecToMind2	0.278	0.387	87.9%	93.4%	88.1%	89.7%	0.769	0.487
Inter	Visual MindBridge	0.091	0.214	75.3%	86.7%	85.2%	80.9%	0.853	0.553
	Visual Umbrae	0.054	0.233	70.7%	85.2%	86.7%	88.2%	0.748	0.482
	Whole ConnecToMind2	0.263	0.407	86.5%	92.6%	86.8%	87.1%	0.771	0.487

Ground Truth	ConnecToMind2	Inter-Subject Umbrae	MindBridge	ConnecToMind2	Intra-Subject Umbrae	MindBridge
						
						
						

Ablation Study for Image Reconstruction

Method	Low-Level				High-Level			
	PCorr ↑	SSIM ↑	Alex(2) ↑	Alex(5) ↑	Incep ↑	CLIP ↑	Eff ↓	SwAV ↓
ConnecToMind2	0.263	0.407	86.5%	92.6%	86.8%	87.1%	0.771	0.487
ConnecToMind2(w/o L)	0.175	0.367	79.5%	88.9%	85.7%	86.2%	0.811	0.517
ConnecToMind2(w/o H)	0.257	0.418	83.5%	90.0%	84.4%	85.8%	0.799	0.511

ConnecToMind2	ConnecToMind2(w/o L)	ConnecToMind2(w/o H)
		
		

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